4 **(a)** A cylinder that contains a fixed amount of ideal gas is shown in **Fig. 4.1.**

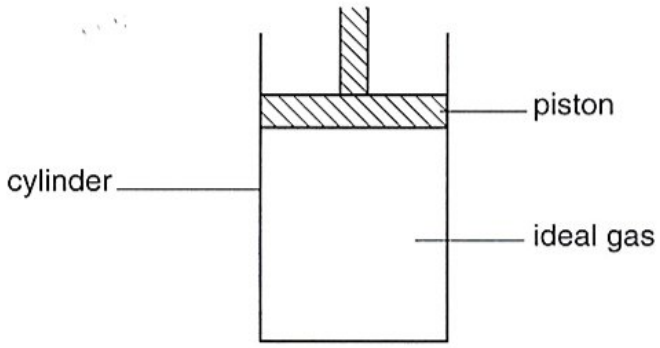


Fig. 4.1

The cylinder has a piston that can move freely.

(i) When the piston is pushed downwards slowly, keeping the temperature constant, explain how the molecules of the gas produce an increase in gas pressure.

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..... [2]

(ii) At one instant, the gas is at a volume of 5.0×10^{-3} m^3, temperature 35 °C and pressure 1.0×10^5 Pa. It is then heated to a temperature of 80 °C at constant pressure.

Determine

1. the volume of the gas at 80 °C,

volume = ...m^3 [2]

2. the work done by the gas.

work done = ... J [1]

(b) An electric kettle, rated 230 V, 8.0 A, contains some water. It is placed on a weighing balance, as shown in Fig. 4.2.



Fig. 4.2

The kettle is switched on and, when the water is boiling, the reading on the balance decreases by 8.1 g in 10.0 s.

(i) Calculate the power rating of the kettle.

power rating = ... W [1]

(ii) Calculate the specific latent heat of vaporisation of water.

specific latent heat = ... J kg^{-1} [2]

(iii) Heat loss to the surroundings is one possible source of error in the determination of the specific latent heat in part **(b)(ii)**. State one other source of error and explain how it affects the calculated value of the specific latent heat.

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..... [2]

[Total: 10]